MTH2010 Problem Sheet 4

During tutorials in Week 9 questions 1, 2, and 5(a) will be discussed.

All rings are commutative with identity $1_R \neq 0_R$.

1. Let R be a ring.

(a) Prove the following identities

(i)
$$0_R a = a 0_R = 0_R$$

(ii)
$$(-a)b = a(-b) = -(ab)$$

(iii)
$$(-a)(-b) = ab$$

(iv)
$$-a = (-1_R)a$$

for all $a, b \in A$.

(v)
$$(-1_R)^2 = 1_R.$$

(b) Prove that if u is a unit of R then $-u$ is a unit.

2. (a) Prove that if R is an integral domain and $r^2 = 1_R$ for some $r \in R$ then either $r = 1_R$ or $r = -1_R$. Show by an example that this is not true if R is not an integral domain.

(b) Prove that if R is an integral domain and $a, b, c \in R$ satisfy $ab = ac$, then either $a = 0_R$ or $b = c$.

3. (a) Let $D < -1$ be a square free integer. Show that $(\mathbb{Z}[\sqrt{D}])^\times = \{1, -1\}$.

(b) Show that in case $D = 3$, or $D = 7$, the group of units of the ring $\mathbb{Z}[\sqrt{D}]$ is infinite.

4. Let f and g be polynomials of degree $n \geq 1$ in $\mathbb{Q}[X]$. Assume that $q_1, q_2, \ldots, q_{n+1}$ are $n+1$ distinct elements of $\mathbb{Q}$ such that $f(q_i) = g(q_i)$, $\forall 1 \leq i \leq n+1$. Show that $f = g$.

5. Let R, S be rings and let $\theta : R \to S$ be a ring homomorphism.

(a) Let r_1, $r_2 \in R$. Prove the following equivalence

$$\theta(r_1) = \theta(r_2) \quad \Longleftrightarrow \quad r_1 - r_2 = r_1 + (-r_2) \in \operatorname{Ker}\theta.$$

(b) Prove that if $u \in R^\times$ is a unit then $\theta(u)$ is a unit and

$$\theta(u)^{-1} = \theta(u^{-1}).$$

6. Prove that the ideals $(2, 1+\sqrt{-5})_R$ and $(3, 2+\sqrt{-5})_R$ of the ring $\mathbb{Z}[\sqrt{-5}]$ are not principal.

7. Let I and J be ideals of a ring R with identity.

(a) Prove that $I + J$ is the smallest ideal of R containing both I and J.

(b) Prove that IJ is contained in $I \cap J$.

(c) Give an example where $IJ \neq I \cap J$.

(d) Prove that if $I + J = R$ then $IJ = I \cap J$.

8. Let

$$\theta : \mathbb{Q}[X] \to \mathbb{Q}(\sqrt{5})$$

be the map defined by

$$\theta(a_0 + a_1X + \cdots + a_nX^n) = a_0 + a_1\sqrt{5} + \cdots + a_n(\sqrt{5})^n.$$

(i) Show that θ is a surjective ring homomorphism.
(ii) Prove that $\operatorname{Ker}\theta = (X^2 - 5)_{\mathbb{Q}[X]}$.
(iii) Deduce that the factor ring $\mathbb{Q}[X]/(X^2 - 5)_{\mathbb{Q}[X]}$ is isomorphic to $\mathbb{Q}(\sqrt{5})$.